Where the Headroom Lies: Block ℓ_p-Norm Statistics for Cyclic Join Bounds

Suan Lee
Semyung University
suanlee@semyung.ac.kr

***Abstract*—Pessimistic cardinality estimators are usually improved by strengthening the *program* that combines degree statistics, from fractional edge covers to LpBound's linear program over the Shannon cone. We ask where the remaining headroom lies for cyclic joins, using LpH-CEG, a fractional block-cover LP over ℓ_p-norm degree sequences of multi-way blocks. On single-relation statistics little is left on Hetionet and IMDb: on a fixed hyperpath the cover LP is provably optimal among weightings valid on every database, and LpBound's polynomial flow LP reaches the polymatroid optimum. Block statistics, degree sequences of two-relation joins conditioned on their interfaces, open the statistics axis. With ten norms on both sides they tighten LpBound by a median 46–920× on 120 registered Hetionet cycles ($|V|$=6–10) under the cover LP, but by only 1.3–6.8× on 120 IMDb cycles, level with the inference gap. The polymatroid program, valid for every p and solved in seconds here, uses them fully: 236–10,200× and 9.7–87×, below LpBound on all 349 Hetionet and IMDb queries; our ℓ_p extension of Im et al.'s flow LP tracks it in polynomial time. On G-CARE (241 AIDS, 192 of 366 YAGO queries) the margin is 15–24×, but under the cover LP the inference gap dominates on AIDS. For a block of two binary relations all statistics but one avoid the join. We prove an ℓ_p Composition Inequality for validity. Limits: statistics come from data slices, their cost grows faster than the data, and bound-guided join orders lose to DuckDB on IMDb.**

***Index Terms*—cardinality estimation, cyclic queries, ℓ_p-norms, fractional cover, degree sequences, pessimistic estimation**

I. Introduction

Cardinality estimation, predicting the output size of a join query before execution, is a central input to query optimization. Errors compound multiplicatively across join operators, and cyclic queries are the hardest case: the constraints imposed by cycle-closing edges break independence assumptions that underpin most estimators.

Pessimistic estimation replaces estimates by guaranteed upper bounds [1]. The AGM bound [2] uses fractional edge covers; PANDA [3] and the degree-based MOLP bound [4] add max-degree (ℓ_∞) constraints; and most recently, **LpBound** [5] generalizes to all ℓ_p-norms of degree sequences, achieving orders-of-magnitude improvements on skewed data via an LP over entropy inequalities. Independently, multi-way *block extensions* have been proposed to capture joint constraints across multiple relations [6], [7].

However, the two lines have not met: *no existing pessimistic bound uses interface-conditioned ℓ_p norms of multi-relation (block) degree sequences* (CorrBound's inner products are an ℓ_1 special case on a shared interface; Section VI). LpBound's program exploits the redundant coverage of cyclic queries, but over single-relation statistics only; block extension methods capture joint constraints, but through max degrees only.

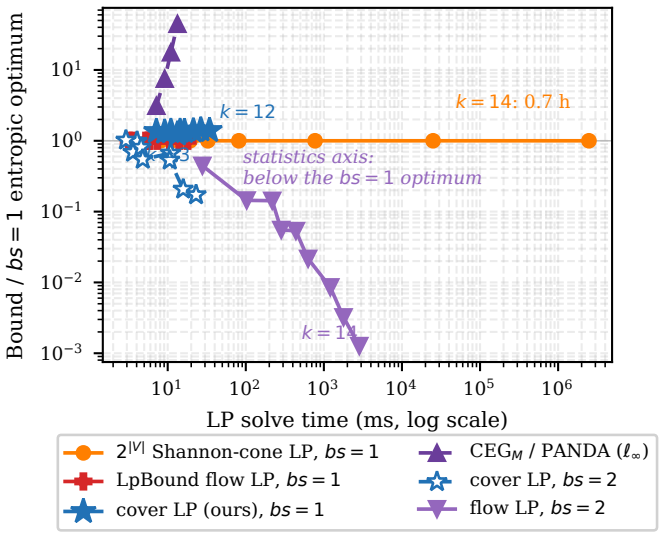


Fig. 1. Both axes on one *measured* cost–tightness plot (each point a synthetic cycle-k query; y is relative to the entropic optimum on single-relation statistics). Above y=1, the inference axis at bs=1: CEG$_M$/PANDA is cheap but up to 45× loose, the cover LP within 1.43× (8–44 ms end to end), and the optimum itself 25 s at k=12 as a $2^{|V|}$ program but a 3–18 ms solve ($k\leq$14) as LpBound's flow LP—that axis is closed. Below y=1, the statistics axis: the same programs on bs=2 block statistics (three norms), the cover LP level with the optimum at $k\leq$4 and up to 6× below it, the flow LP (Section V-E) 2–800× below.

a) Our contribution: LpH-CEG bridges ℓ_p-norm LP optimization with multi-way block decompositions through a *fractional block cover LP*: on cyclic queries an attribute is covered by several steps, and the slack admits fractional weights $w_j < 1$ that, with intermediate norms (ℓ_2), tighten the max-degree bound (w=1, p=∞) dramatically. LpBound's entropic LP over the Shannon cone is related precisely in Section IV-A. Figure 1 places the bs=1 configuration in the design space: near the entropic optimum and decomposable into per-step residuals, but, since LpBound's flow LP reaches that optimum in polynomial time, not cheaper in any sense that matters. The thesis is at the other end: at $bs\geq$2 the statistics cost more to collect, once per database, and the bound is far tighter (Section V-C).

Our contributions are:

1) **On single-relation statistics the inference axis is**

closed (Section IV-A). On a fixed hyperpath the cover LP is optimal over *every* weight vector valid on all databases (Proposition 3). The gap to the entropic optimum comes from statistics the hyperpath leaves unused, not from the weighting—and LpBound's flow LP, which uses them, reaches that optimum in polynomial time, so at $bs{=}1$ the cover LP is dominated as a bound.

2) **The statistics axis is open (Section V-C).** Block statistics $\mu_p(F \mid I)$, $|F| \geq 2$, are a median $10.7\times$ below LpBound with the same ten norms on the 109 short cycles of a registered Hetionet suite (a subset and 3K-row prefixes chosen after registration), and 46–$920\times$ on $120/120$ longer Hetionet cycles ($|V|{=}6$–10), against a median 2.7–$3.1\times$ of inference headroom. On 120 IMDb cycles from JOB's snapshot the cover-LP margin is only 1.3–$6.8\times$ against a 5.1–$6.8\times$ inference gap (above the entropic bound on 8 queries); the polymatroid program's is 9.7–$87\times$ on $120/120$. On the external G-CARE benchmark (433 cyclic AIDS and YAGO queries run) it is 15–$24\times$, while the cover LP's is far below the inference gap on AIDS ($1.35\times$ against $59\times$) and comparable to it on YAGO ($8.9\times$ against $6.1\times$). On homogeneous synthetic cycles the axes are comparable.
3) **Richer statistics can reopen the program (Sections V-C, V-E).** On the same block statistics the polymatroid optimum beats the cover LP by 1.6–$3.8\times$ ($12/12$ synthetic) and a median 5.0–$11.1\times$ (Hetionet) and 7.6–$11.6\times$ (IMDb), against 1.2–$1.5\times$ (synthetic), 2.7–$3.1\times$ (Hetionet) and 5.1–$6.8\times$ (IMDb) on single-relation statistics. The $2^{|V|}$ program over the $bs{\leq}2$ catalog solves every real-data query in at most 63 s; our ℓ_p extension of Im et al.'s flow LP matches it on 633 of 862 queries (within $2.41\times$) in polynomial time and scales to $k{=}16$ (4.4 s).
4) **Which block statistic matters (Section V-C).** Under the same program (the order-relaxed flow LP, so a lower bound on the gain), block statistics carry a further median $63\times$ on the Hetionet long suite beyond pair-join sizes (three norms; 19–$154\times$ by $|V|$), a proxy for the two-relation statistic CorrBound adds (itself worth $31\times$ there; on IMDb, where half the pairs are self-joins, sizes add almost nothing). For a block of two binary relations all but one of them come from single-relation degree tables without the pair join; the exception, the ℓ_2 norm of the common-neighbour counts, restores the full bound on $85/120$ Hetionet and $116/120$ IMDb queries.
5) **Validity, cost and limits (Sections IV, V, V-F).** An ℓ_p Composition Inequality makes the cover bound valid for every hyperpath, block size and p; the proof turns redundant coverage into a fractional cover, and 207K databases plus an LP certificate over 6,197 hyperpath structures independently agree. On Hetionet's 20K-row slices block statistics cost a median $61\times$ more to collect and, once a workload catalog saturates, 10 ms per estimate; collection grows faster than the data (up to 25 min and 16 GiB per query at 1,000 IMDb films), and the statistics are collected without predicates. Guiding join order, the bound trims the catastrophic tail on 24 small synthetic cyclic queries (P95 slowdown $1.2\times$ vs. $5.8\times$) rather than the average, and on IMDb block statistics pick the exact-count order on $24/40$ queries, yet those left-deep plans still lose to DuckDB's own; and a per-step residual ordering heuristic does not generalize (median -1.3% over 66 queries).

II. Preliminaries

A. Query Hypergraphs

A conjunctive query $\mathcal{Q}$ over relations $R_1, \ldots, R_m$ with attributes $V = \{a_1, \ldots, a_n\}$ is represented as a hypergraph $H = (V, \mathcal{E})$ where each hyperedge $e_i \in \mathcal{E}$ corresponds to the attribute set of relation R_i. The full natural join is $\mathcal{Q} = \bowtie_{i=1}^{m} R_i$.

B. CEG and MOLP

Cardinality Estimation Graphs [7], [8] model both optimistic and pessimistic estimators as path-finding in a directed graph. The pessimistic CEG_M uses max-degree edge weights: for relation R with interface I, $\mu(R \mid I) = \max_{a_I} |\text{Ext}_R(a_I)|$. The MOLP bound [4] equals the minimum-weight path in CEG_M.

C. Block Extensions and Hyperpaths

A *block* $F \subseteq \mathcal{E}$ is a connected subset of relations. The *block extension factor* [6] is $\mu(F \mid I) = \max_{a_I} |\text{Ext}_F(a_I)|$, where $I = \text{attr}(S) \cap \text{attr}(F)$ is the interface with the current state. A *hyperpath* $P = (S_0{=}\emptyset, F_1, S_1, \ldots, F_m, S_m{=}\mathcal{E})$ decomposes the query into sequential block extensions. The product $\prod_{j=1}^{m} \mu(F_j \mid I_j)$ is a valid cardinality upper bound (the unit-weight ℓ_∞ case of Theorem 1; also [6]). Our framework generalizes this product by replacing ℓ_∞ norms with LP-optimized ℓ_p-norms.

D. LpBound: ℓ_p-Norm Degree Sequences

LpBound [5] uses ℓ_p-norms of degree sequences as tighter statistics. For relation R with degree sequence $\mathbf{d} = (d_1, \ldots, d_k)$ where $d_i = |\text{Ext}_R(a_i)|$, the ℓ_p-norm is $\|\mathbf{d}\|_p = (\sum_i d_i^p)^{1/p}$. The core inequality is: for any $p \geq 1$,

$$\tfrac{1}{p}\, h(X) + h(Y \mid X) \leq \log \|\mathbf{d}\|_p$$

for the degree sequence $\mathbf{d}$ of Y given X in R. Optimizing over all valid inequalities via an LP yields bounds orders of magnitude tighter than max-degree alone.

III. LpH-CEG Framework

A. ℓ_p-Norm Block Extension Factor

Definition 1 (ℓ_p Block Extension Factor)**.** *For a block F with interface I, let $\mathbf{d} = (d_1, \ldots, d_k)$ be the degree sequence where $d_i = |\text{Ext}_F(a_i)|$ for each distinct interface value $a_i \in \pi_I(\mathcal{Q}_F)$. The ℓ_p block extension factor is:*

$$\mu_p(F \mid I) = \|\mathbf{d}\|_p = \Big(\sum_{i=1}^{k} d_i^p\Big)^{1/p}$$

Algorithm 1 LpH-CEG Estimation

Require: Query hypergraph H, relation data D, block size limit b, p-values $\mathcal{P} = \{p_1, \ldots, p_r\}$
Ensure: Upper bound $\hat{C}$ on $|\mathcal{Q}(D)|$
1: Enumerate blocks $\mathcal{B} \leftarrow \{F \subseteq \mathcal{E} : |F| \le b,\ F \text{ connected}\}$
2: **for** each block $F \in \mathcal{B}$ and interface I **do**
3: Compute join $Q_F \leftarrow \bowtie_{R \in F} R$
4: Collect degree sequence $\mathbf{d}_{F,I}$
5: Compute $\mu_p(F \mid I)$ for each $p \in \mathcal{P}$
6: **end for**
7: Build block-extension DAG with ℓ_∞ weights
8: Find K shortest hyperpaths $P_1, \ldots, P_K$ $\{K{=}1$: default$\}$
9: **for** each hyperpath $P_i = (F_1, \ldots, F_m)$ **do**
10: Solve LP (1)–(2) for weights $(w^*_{j,p})$
11: $\hat{C}_i \leftarrow \prod_{j,p} \mu_p(F_j \mid I_j)^{\,w^*_{j,p}}$
12: **end for**
13: **return** $\min(\hat{C}_1, \ldots, \hat{C}_K,$ max-deg bound$)$

Special cases: $\mu_\infty(F \mid I) = \max_i d_i = \mu(F \mid I)$ (max-degree); $\mu_1(F \mid I) = \sum_i d_i = |\mathcal{Q}_F|$ (cardinality).

B. Fractional Block Cover LP

Given a hyperpath P with m steps, we define a variable $w_{j,p} \ge 0$ for each step j and each p-value. The LP minimizes the log-space bound:

$$\min \sum_{j=1}^{m} \sum_{p \in \mathcal{P}} w_{j,p} \cdot \log_2 \mu_p(F_j \mid I_j) \tag{1}$$

subject to the fractional block cover constraints: for each attribute $a \in V$,

$$\sum_{p \in \mathcal{P}} \left(\sum_{j:\, a \in \text{attr}(F_j) \setminus I_j} w_{j,p} \;+\; \sum_{j:\, a \in I_j} \frac{w_{j,p}}{p} \right) \;\ge\; 1 \tag{2}$$

The LP has $O(m \cdot |\mathcal{P}|)$ variables and $|V|$ constraints, solvable in about 1–2 ms for typical queries.

C. Why Cyclic Queries Benefit

a) Where the slack comes from: With binary extensions every attribute is new at exactly one step. On a star no attribute reappears in a later interface, so the LP returns $w_j{=}1$ and the max-degree bound. On a path an interior attribute is re-covered by the next interface at rate $1/p$, so the LP may trade an extension for an interface at $p{=}1$ (on a 4-path it returns $(1, 0, 1)$, an AGM-style $N_1 N_3$)—a gain from cardinalities, not from skew. On a cycle the closing step re-covers attributes already bound, so every step's constraint has slack and intermediate norms enter; this is the source of Theorem 4.

D. Algorithm

Algorithm 1 summarizes the LpH-CEG estimation pipeline.

The LP in line 10 has $O(m \cdot |\mathcal{P}|)$ variables ($m \le |\mathcal{E}|$ steps, $|\mathcal{P}| = 3$ in practice) and n constraints, solved by HiGHS [9] in about 1–2 ms. Line 13 ensures LpH-CEG never produces a looser bound than the standard max-degree bound.

IV. Theoretical Results

Proofs not given in full here are in the extended version shipped with the artifact (code, data pipeline, results: `github.com/suanlab/LpH-CEG`).

Theorem 1 (ℓ_p Composition Inequality)**.** *Let $P = (S_0{=}\emptyset, \ldots, S_m{=}\mathcal{E})$ be a hyperpath, $\mathcal{P} \subseteq [1, \infty]$ (with $1/\infty{=}0$), and let $(w_{j,p})$ satisfy the cover constraint* (2)*, with the statistics computed exactly on the database D. Then*

$$|\mathcal{Q}| \le \prod_{j=1}^{m} \prod_{p \in \mathcal{P}} \mu_p(F_j \mid I_j)^{\,w_{j,p}}$$

Proof. Let h be the entropy of the uniform distribution on $\mathcal{Q}(D)$, so $h(V) = \log_2 |\mathcal{Q}|$. Write $A_j = \text{attr}(F_j)$, $V_j = \text{attr}(S_j)$ and $E_j = A_j \setminus I_j$: the E_j partition V and $I_j \subseteq V_{j-1}$. Since $\pi_{A_j} \mathcal{Q} \subseteq \bowtie_{R \in F_j} R$, the ℓ_p inequality of [10] gives $\frac{1}{p} h(I_j) + h(E_j \mid I_j) \le \log_2 \mu_p(F_j \mid I_j)$. Weight by $w_{j,p}$ and sum; with $w_j = \sum_p w_{j,p}$ and $s_j = \sum_p w_{j,p}/p$ it remains to show $\sum_j [s_j h(I_j) + w_j h(E_j \mid I_j)] \ge h(V)$, where (2) reads $w_{j(a)} + \sum_{j' : a \in I_{j'}} s_{j'} \ge 1$ for the step $j(a)$ that binds a. Let $g_j = h(E_j \mid V_{j-1})$ and $\delta_j = \max(0, 1 - w_j)$. (i) By the chain rule and submodularity, $h(I_{j'}) \ge \sum_j h(I_{j'} \cap E_j \mid V_{j-1})$. (ii) If $\delta_j > 0$, (2) makes $\{I_{j'} \cap E_j\}$, weighted $s_{j'}/\delta_j$, a fractional cover of E_j; $S \mapsto h(S \mid V_{j-1})$ is monotone submodular, hence fractionally subadditive, so $\sum_{j'} s_{j'} h(I_{j'} \cap E_j \mid V_{j-1}) \ge \delta_j g_j$. (iii) As $I_j \subseteq V_{j-1}$, $w_j h(E_j \mid I_j) \ge (1 - \delta_j) g_j$. Summing gives at least $\sum_j g_j = h(V)$. □

Only monotonicity and submodularity are used, so the argument holds over the whole polymatroid cone: the inequality is a Shannon-flow inequality [3], [11], here proved constructively for this family. The redundant coverage that lets per-step weights fall below 1 (the source of every cyclic improvement, and the reason the elementary per-step argument ($w_j \ge 1$) does not apply) enters exactly as the fractional cover in (ii). The theorem assumes exact statistics of the database being bounded; the sketched ℓ_2 of Section V-C is outside it.

a) Independent checks: Two checks that do not rely on the proof agree with it. Over 207,000 databases (every database over a two-value domain for the triangle and 4-cycle, plus 15,000 random ones across six shapes at block sizes 1–3) the estimate never fell below the exact count (minimum ratio 1.0). And because the remaining inequality is linear in h and affine in w, an LP over the polymatroid cone decides it for every vertex of the feasible weight region: across 6,197 hyperpath structures on eleven shapes with $|V| \le 6$ and block sizes 1–3 (1.73M vertices; every hyperpath on 19 of the 33 (shape, block size) pairs) the minimum slack is exactly 0—tight, never negative. Both ship as regression gates.

Theorem 2 (Subsumption)**.** *LpH-CEG subsumes:*

1) *Max-degree block bounds [6]: setting $p = \infty$, $w_j = 1$ recovers the Composition Lemma.*
2) *Binary ℓ_p-cover bounds: setting $|F_j| = 1$ gives LP* (1)–(2) *at bs=1—a polynomial-size relaxation of Lp-Bound's [5] entropic LP (Section IV-A), not necessarily its entropic optimum.*

3) *AGM bound [2]: with* $|F_j|{=}1$ *and* $p{=}1$ *on every step makes each attribute's coverage* 1 *per covering step, so* (2) *becomes the fractional edge-cover constraint and the objective becomes* $\prod_j |R_j|^{w_j}$.

A. Relationship to LpBound

LpBound [5] also applies to cyclic queries and also solves a linear program over ℓ_p-norm degree statistics, so we must delineate precisely what the block-cover LP adds.

a) Two different LPs: LpBound's bound is the optimum of an *entropic* LP: it maximizes $h(V)$ over the Shannon cone (monotonicity and submodularity inequalities on all $2^{|V|}$ entropic terms) subject to the ℓ_p-norm statistics constraints [5], [10]. We follow LpBound in calling it the entropic bound, although the Shannon cone is the polymatroid relaxation of the entropic cone, so it is the strongest bound LpBound computes from those statistics, not necessarily the tightest they admit. Our LP (1)–(2) instead optimizes $O(m\,|\mathcal{P}|)$ weights over a *fixed hyperpath*, with one fractional-cover constraint per attribute. On identical statistics the entropic optimum is the tighter of the two by construction, and we do not claim otherwise: at $bs{=}1$ the cover bound is 1.25–$1.43\times$ looser on synthetic power-law cycles (Figure 2), 1.9–$3.5\times$ on six hand-picked real Hetionet cycles, and up to $8.8\times$ on one relation repeated along a long chain, where the entropic cone exploits conditional structure the fixed hyperpath treats independently.

b) On a fixed hyperpath, reweighting is exhausted: Could a better weighting of the hyperpath's statistics close that gap? No: the cover constraint is also necessary.

Proposition 3 (Necessity)**.** *If nonnegative weights* $(w_{j,p})$ *make the product of Theorem 1 an upper bound on* $|\mathcal{Q}(D)|$ *for every database* D*, they satisfy* (2)*. So on a fixed hyperpath the LP* (1)–(2) *is optimal over all weightings of its statistics that are valid on every database.*

Proof. Fix $a \in V$ and $N \geq 2$, and let D_N give a the domain $[N]$ and every other attribute a single value. Then $|\mathcal{Q}(D_N)| = N$, and $\mu_p(F_j \mid I_j)$ is $N^{1/p}$ if $a \in I_j$, N if $a \in A_j \setminus I_j$, and 1 otherwise. The bound is N^{c_a}, with c_a the left side of (2), and $N^{c_a} \geq N$ forces $c_a \geq 1$. □

(In entropy terms, D_N is the modular polymatroid $h(S) = [a \in S] \log_2 N$.) A cutting-plane search with the polymatroid LP confirms it on triangle, cycle-4/5/6 and K_4 (cycle-4: 15.335 in $\log_2$, against 14.858 for the polymatroid optimum). The residual gap comes from *statistics* the hyperpath leaves unused, which the polymatroid program exploits and Section V-C extends.

c) Efficiency: The Shannon-cone *formulation* of that optimum has $2^{|V|}$ variables and $O(2^{|V|}|V|^2)$ constraints and degrades accordingly: 81 ms at $|V|{=}8$, 25 s at $|V|{=}12$, 41 min at $|V|{=}14$, unsolvable beyond (Figure 2; the same curve on real Hetionet degree sequences: 22 s and 27 min). But that is not what LpBound solves. Its LPflow [5, Thm. 4.5] rewrites every simple ℓ_p constraint (one conditioning attribute) as a flow bundle after Im et al. [12] and is polynomial; our reimplementation returns the entropic bound to 10^{-6} on every instance we compared (12 family–seed pairs, the sweep, the 240 long-suite queries) in 3–26 ms at $k{=}3$–16 (Figure 2). So at $bs{=}1$ the cover LP buys *no* asymptotic advantage over LpBound: as a $bs{=}1$ bound it is dominated (looser, and both solve in milliseconds). The fixed hyperpath buys the per-step decomposition (item 1) and the framework the statistics vocabulary (item 2), where inference becomes non-trivial again (Section V-E).

d) What the block-cover LP adds: What LpBound's single global program does not provide is structural:

1) **A per-step residual bound.** Because the LP is attached to a hyperpath, it yields a bound *per extension step*, and hence a residual bound for any variable prefix. This is exactly the signal our WCOJ variable ordering consumes (Section V-F); LpBound's single global optimum exposes no such per-step decomposition.
2) **Block-level statistics.** With multi-way blocks ($|F_j| > 1$) the LP draws on ℓ_p-norms of *joined* block relations—statistics outside LpBound's single-relation vocabulary.

So at $bs{=}1$ LpH-CEG trades a little tightness for a decomposable bound; item 2 is the axis the paper is about (Section V-C).

Theorem 4 (Cyclic Improvement)**.** *Consider a cycle query* C_k *(*$k \geq 3$*) evaluated by binary extensions in cycle order, where relation* R_j *has cardinality* $N_j = |R_j|$*, maximum degree* M_j*, and* ℓ_p*-norms* $\mu_p(R_j)$ *of its degree sequence from* a_j *to* a_{j+1}*. Then:*

1) ***(Standard bound.)*** *The max-degree bound is*

$$B_{\max} = N_1 \prod_{j=2}^{k-1} M_j,$$

where the first extension has empty interface and contributes the cardinality N_1*, the* $k-2$ middle *extensions each contribute a max-degree* M_j*, and the cycle-closing extension (interface of size two, empty extension set) contributes a factor of* 1.

2) ***(LP bound.)*** *For every* $p \in \mathcal{P}$ *(with* $1 \in \mathcal{P}$*) the fractional block-cover LP* (1)–(2) *is feasible with weight* $p/(p{+}1)$ *and exponent* p *on every middle extension, together with the split* $(\frac{p}{p+1}, \frac{1}{p+1})$ *of unit weight across the first and closing* (ℓ_1) *extensions. This yields*

$$B_{\mathrm{LP}} \leq \min_{p \in \mathcal{P}} \; N_1^{\frac{p}{p+1}} N_k^{\frac{1}{p+1}} \prod_{j=2}^{k-1} \mu_p(R_j)^{\frac{p}{p+1}},$$

e.g. $N_1^{2/3} N_k^{1/3} \prod_j \mu_2(R_j)^{2/3}$ *at* $p{=}2$*. (The exponent must be common, or non-increasing along the cycle: the cover constraint of the attribute between edges* j *and* $j{+}1$ *reads* $\frac{p_j}{p_j+1} + \frac{1}{p_{j+1}+1} \geq 1$*.)*

3) ***(Improvement.)*** *Therefore, for every* $p \in \mathcal{P}$*,*

$$\frac{B_{\max}}{B_{\mathrm{LP}}} \;\geq\; \Big(\frac{N_1}{N_k}\Big)^{\frac{1}{p+1}} \prod_{j=2}^{k-1} \frac{M_j}{\mu_p(R_j)^{\,p/(p+1)}}.$$

At p=2 a middle factor exceeds 1 iff $\mu_2(R_j)^2 < M_j^3$: the sum of squared degrees is below the cube of the maximum degree, as when a few high-degree values carry the relation (n values of uniform degree d qualify iff $n < d$; many degree-1 values do not). As $p \to \infty$ every factor tends to 1. The boundary factor is a constant independent of k; for a uniform *cycle ($N_j \equiv N$) it equals 1, and in the general case it is ≥ 1 for a hyperpath entering the cycle at the largest relation ($N_1 \geq N_k$). If every middle factor is at least $c > 1$, the ratio is at least $c^{k-2}(N_1/N_k)^{1/(p+1)}$: it grows geometrically in the number of middle edges. Since LpH-CEG returns $\min(B_{\max}, B_{\mathrm{LP}})$, the realized ratio is always ≥ 1.*

Proof sketch. The standard bound (1) follows directly from the cycle hyperpath structure: the first extension has empty interface, so its degree sequence is the single value N_1 and $\mu_\infty = N_1$; each middle extension adds one attribute over a size-one interface with $\mu_\infty = M_j$; the closing extension has both endpoints already in the state, so its conditional degrees are all 1 and $\mu_\infty = 1$. For (2), each attribute is covered as an extension in one step (rate 1) and as an interface in the next (rate $1/p$); with a common p the local constraint $w + w/p \geq 1$ is tight at $w = p/(p+1)$. Attribute a_2 then reads $w_1 + \frac{1}{p+1} \geq 1$, a_k reads $\frac{p}{p+1} + w_k \geq 1$ and a_1 reads $w_1 + w_k \geq 1$, all tight at $w_1 = \frac{p}{p+1}$, $w_k = \frac{1}{p+1}$. The closing step has $\mu_1 = N_k$, so the first/closing contributions are $N_1^{p/(p+1)} N_k^{1/(p+1)}$ against N_1 in $B_{\max}$. The LP optimum is at most every such point, and may mix p further. □

e) Observation: On power-law cycles with fixed parameters the middle factors exceed 1, so the bound predicts a ratio geometric in k; the measured one grows accordingly (10-seed means): 8.4× (k=4), 69× (k=6), 614× (k=8).

B. Tightness

Theorem 1 establishes that LpH-CEG bounds are *valid* (never underestimate), but not necessarily *tight* (matching a worst-case instance). Tightness depends on the degree sequence structure:

Each step's ℓ_p inequality is attained for any interface (by the distribution $P(u) \propto d_u^p$ with a uniform conditional).

Per-step tightness does not make the *composition* tight: the weighted product can be loose even where every factor is exact, and we measure it to be 1.25–1.43× looser than the entropic optimum at bs=1 on synthetic cycles and a median 2.7–6.8× on real data, 59× on G-CARE's AIDS (Section IV-A).

This creates a principled trade-off: binary extensions keep simple interfaces; multi-way blocks capture block-level correlation at the cost of non-simple ones. Results use bs=1 unless marked bs=2.

C. Delta over the Block Framework

We inherit the block-extension DAG and hyperpath machinery from H-CEG [6], which uses only ℓ_∞ weights with no optimization layer. What this paper adds is the ℓ_p fractional cover LP that replaces those fixed scalars, the separation of the statistics and inference axes (Section V-C), the flow-LP inference over block statistics (Section V-E), and the per-step residual signal; none exists in H-CEG. Theorem 4 additionally supplies a structural test neither predecessor provides, predicting the magnitude from a product of $k-2$ per-edge factors $M_j/\mu_p(R_j)^{p/(p+1)}$ at a common p.

V. Experimental Evaluation

A. Setup

a) Datasets: We evaluate on five real-world datasets: **Hetionet** [13] (biomedical knowledge graph, 45K nodes, 2M edges, heterogeneous); **Epinions** (social network, 75K nodes, 509K edges); **IMDb**, JOB's snapshot [14]: the 200 most-voted feature films for a cycle suite, and the JOB-medium subset; and the AIDS and YAGO graphs of G-CARE [15] with its cyclic queries. We also use synthetic power-law graphs (100–5,000 nodes, 400–20,000 edges). Seeds and runs are stated per table: multi-seed results are 10-seed means; single-instance tables say so.

b) Queries: Synthetic cycle-k for $k \in \{3, \ldots, 16\}$ plus diamond and K_4 families; on real data, three registered suites (Section V-D) plus a handful of hand-picked cycles.

c) Baselines: AGM (fractional edge cover, $\prod |R_i|^{w_i}$); CEG$_M$ (binary max-degree shortest path, bs=1; equivalent to PANDA/MOLP on binary queries [3], [4]); LpH-CEG (LP-optimized ℓ_p-norms, $\mathcal{P} = \{1, 2, \infty\}$). Every program in a comparison gets the same $\mathcal{P}$, including our entropic LpBound. Because the entropic bound gains more from extra norms than the cover LP does, the comparisons against LpBound on real data (the abstract, Table IV's upper block, the short suite and Table III) use $\mathcal{P} = \{1, \ldots, 10, \infty\}$ on both sides; every other comparison with the entropic bound (the synthetic tables and figures, the six hand-picked cycles, the CorrBound proxy, the gap predictor) uses three norms, $\{1, 2, \infty\}$. The "LpH-CEG" column throughout denotes our *cover*-LP at bs=1 (Section IV-A); it is *not* the entropic LpBound of Zhang et al. [5], against which we compare separately in Section IV-A (our cover bound is 1.25–1.43× looser on synthetic cycles, a median 2.7–6.8× on Hetionet and IMDb). Learned and summary estimators (FactorJoin, GNCE, COLOR [16]–[18]) produce *point* estimates without guaranteed upper bounds; CorrBound, which enriches the entropic program with cross-relation statistics, is compared through a proxy in Section V-C.

d) Unified bound comparison: Table I places all four guaranteed-bound methods on one suite. The ordering is as predicted: CEG$_M$ $\geq$ Cover-LP $\geq$ Entropic, with AGM incomparable (it ignores degrees). Our cover-LP is 2.3–35.7× tighter than CEG$_M$ on synthetic cycles, is also consistently tighter than AGM (e.g. 2.0× on cycle-6), and stays within 1.25–1.36× of the entropic (polymatroid) optimum on the same single-relation ℓ_p statistics on these cycle-3–6 instances.

TABLE I

UNIFIED COMPARISON OF GUARANTEED UPPER BOUNDS (SYNTHETIC POWER-LAW CYCLES, 100V/400E, ONE INSTANCE). "COVER" IS OUR bs=1 LP; "ENTROPIC" IS THE $2^{|V|}$ LPBOUND. LOWER IS TIGHTER. "/CEG" AND "/ENT" ARE COVER'S RATIO TO CEG_M AND TO THE ENTROPIC OPTIMUM.

Query	AGM	CEG_M	Cover	Entropic	/CEG	/Ent
Cycle-3	8.0K	15.6K	6.8K	5.0K	2.3×	1.36×
Cycle-4	160K	655K	116K	87K	5.7×	1.32×
Cycle-5	3.2M	25.5M	1.97M	1.45M	13.0×	1.36×
Cycle-6	64M	1.12B	31.5M	25.2M	35.7×	1.25×

TABLE II

LPH-CEG VS. CEG_M ON SYNTHETIC CYCLE QUERIES (bs=1, $|V|$=200, $|E|$=800, MEAN ± STD OVER 10 SEEDS). THE RATIO IS A *bound-tightening* MEASURE (CEG_M/LPH-CEG); THE q-ERROR COLUMN (ACCURACY VS. THE TRUE COUNT) IS SHOWN ONLY WHERE THE EXACT COUNT IS FEASIBLE TO MATERIALIZE, AND IS "—" FOR $k \geq 6$ WHERE THE TRUE CARDINALITY IS INTRACTABLE TO COMPUTE.

Query	CEG_M	LpH-CEG	Ratio	q-error
Cycle-3	53K±2K	19K±0.3K	2.75±0.13×	10.8±0.2×
Cycle-4	3.8M±253K	448K±44K	8.43±0.78×	18.3±1.8×
Cycle-5	270M±18M	11.3M±308K	23.8±1.3×	39.0±1.1×
Cycle-6	18.8B±2.3B	272M±16M	68.9±6.4×	—
Cycle-7	1.4T±167B	6.5B±329M	211±20×	—
Cycle-8	97T±14T	158B±10B	**614±71**×	—

B. Main Results: Cyclic Query Bounds

Table II shows the core result: LpH-CEG achieves 2.8× to 614× tighter bounds than CEG_M on cyclic queries, with tight standard deviations (relative std $< 12\%$) confirming stability across random graph instances. The improvement *scales with cycle size*: from 8.4× on cycle-4 to 69× on cycle-6 to 614× on cycle-8. It also *scales with data size*: cycle-4 improves from 5.9× (100V) to 34× (2KV), with the dependence on skew characterized in Section V-G.

C. Statistics Versus Inference

The two axes can be separated directly: hold the program fixed and change the statistics, then hold the statistics fixed and change the program. Table III is that 2×2, over four families and three seeds, with the exact join size computed so every bound is checked rather than assumed.

Reading the table by column pairs gives the central observation. At fixed program, the statistics axis (single → block) improves the bound by 1.5–2.7× for the cover LP and 1.8–5.9× for the optimum; at fixed statistics, the inference axis (cover → optimum) improves it by 1.2–1.5× on single-relation statistics and 1.6–3.8× on block ones, over all 12 pairs (three norms give the same picture: 1.5–2.3, 1.7–5.5, 1.15–1.5, 1.5–3.8). The axes interact: richer statistics give the program more to exploit. The statistics axis has room because a block statistic $\mu_p(F \mid I)$ with $|F| \geq 2$ describes a multi-relation join, which no single-relation degree *sequence* determines (most of it follows from single-relation degree *tables*; see below).

Two scoping points keep this from being overstated, both measured on the headline cycle protocol ($|V|$=200, $|E|$=800, 10 seeds; ratios of seed means over a different seed set from Table II's, one of ten shared, so cycle-8 reads 654× here and 614× there) so the axes are in the same units:

TABLE III

STATISTICS × INFERENCE, 40 VERTICES / 120 EDGES, SEED 101 (12 FAMILY–SEED PAIRS BEHAVE ALIKE; SEE TEXT). BOTH PROGRAMS SEE THE SAME CATALOG (EVERY SINGLE RELATION, OR EVERY $bs \leq 2$ BLOCK AND INTERFACE) WITH THE SAME TEN NORMS. EVERY ENTRY IS A VALID UPPER BOUND ON THE EXACT COUNT.

	cover LP		polymatroid optimum	
Query	single	block	single	block
Triangle	1,083	668	806	**352**
Cycle-4	9,855	4,716	7,388	**1,355**
Diamond	9,769	4,281	7,388	**1,252**
K_4	9,125	3,376	6,320	**1,093**

Axis	cycle-4	cycle-6	cycle-8
inference: max-deg → cover LP	8.8×	72.7×	654×
inference: cover LP → entropic	1.25–1.43× (3 or 10 norms)		
statistics: bs=1 → bs=2	1.21×	1.87×	2.70×

First, the claim concerns headroom *remaining once an LP is in place*. The first inference step, from a max-degree product to the cover LP, dwarfs both of the others and is the contribution of the LP itself. Second, the separation between the two remaining axes is regime-dependent: on these homogeneous synthetic cycles they are comparable (1.2–2.7× against 1.25–1.43×), and it is heterogeneous real data that separates them, where on the 109-query short suite the statistics step (cover bs=1 → bs=2) is 30× against an inference gap of 2.8× with ten norms (56× against 2.6× with three).

The diagonal comparison is the practical one. The *cheap* LP on rich statistics (4,716) beats the *optimal* LP on poor statistics (7,388)—on 12/12 family–seed pairs, and on all 109 cycle-3/4 queries of the registered Hetionet suite, where the median ratio to the entropic bound is 0.094× (0.057× with three norms). That subset and its 3K-row file-order prefix (to keep pair joins small) were chosen after registration, and cycle-5 was not run at bs=2. All 109 bounds were checked against exact counts with no violation, but the counts are only 0–334 (median 30) and bs=2 is still a median 85× above them: the ordering moved, not the distance to the truth—Table IV is the uncapped test. The claim does *not* say the cover LP is a better program. At equal statistics the optimum wins on 12/12 pairs, as it must. It says that the choice of statistics comes first, and it identifies the strongest cell—the optimum on block statistics, i.e. LpBound run on materialized pair-join views, which no published evaluation does; Section V-E computes it in polynomial time.

Table IV carries the comparison to the sizes the introduction motivates, $|V|$=6–10, on two registered suites (Section V-D). With ten norms on both sides the two axes diverge with $|V|$ on Hetionet: the inference gap stays at 2.7–3.1×, while bs=2 statistics are 46/270/920× tighter than the entropic bound at the median under the cover LP and 236/1,720/10,200× under the polymatroid program, on 120/120 queries. The

TABLE IV

TWO REGISTERED REAL SUITES (IMDB AFTER A DISCLOSED PILOT), 40 CYCLES PER LENGTH ADMITTED IN SHA-256 ORDER OF THE CANONICAL RELATION SEQUENCE. HETIONET: METAGRAPH CYCLES OVER DISTINCT RELATIONS, THE FIRST 20K ROWS OF EACH RELATION IN FILE ORDER. IMDB (JOB SNAPSHOT): THE 200 MOST-VOTED FEATURE FILMS (A DENSE, POPULARITY-BIASED SLICE), NINE CAST ROLES, TWO COMPANY TYPES AND KEYWORDS, ONE RELATION EACH. MEDIANS OVER QUERIES; "POLYMATROID" IS THE $2^{|V|}$ PROGRAM AND "FLOW" OUR FLOW LP, BOTH OVER THE $bs{\leq}2$ CATALOG. UPPER BLOCK: EVERY PROGRAM WITH TEN NORMS ($\mathcal{P} = \{1, \ldots, 10, \infty\}$), THE COMPARISON AGAINST LPBOUND; LOWER BLOCK: THREE ($\{1, 2, \infty\}$, THE COVER LP'S DEFAULT). EXACT COUNTS BY DUCKDB (60 S BUDGET); QUERIES THAT TIME OUT ARE UNCHECKED, NOT DROPPED, SO THE "/EXACT" ROWS ARE MEDIANS OVER 40/40/35 AND 40/33/28 QUERIES. IMDB TIMES WERE TAKEN UNDER 10 CONCURRENT WORKERS. THE CATALOGED ROW AMORTIZES BLOCK BUILDS OVER THE WORKLOAD; THE $|V|$=6 QUERIES, RUN FIRST, PAY MOST OF THEM.

	Hetionet			IMDb		
$\lvert V\rvert$	6	8	10	6	8	10
Ten norms on both sides:						
cover bs=1 / entropic	2.66×	3.10×	3.00×	5.14×	6.75×	5.96×
cover bs=2 / entropic	0.022×	0.0037×	0.0011×	0.79×	0.35×	0.15×
polymatroid bs=2 / entropic	0.0042×	0.00058×	0.000098×	0.10×	0.034×	0.012×
of 40 below entropic, cover; polym.	40; 40	40; 40	40; 40	32; 40	40; 40	40; 40
entropic / exact	$8.4{\cdot}10^3$	$3.6{\cdot}10^5$	$2.8{\cdot}10^7$	23	171	$1.3{\cdot}10^3$
polymatroid bs=2 / exact	47	277	$2.8{\cdot}10^3$	2.2	6.4	15
Three norms on both sides:						
max-degree / cover bs=1	1.8×	5.1×	8.9×	1.8×	1.9×	3.2×
cover bs=1 / entropic	2.53×	3.24×	3.61×	3.75×	4.33×	4.14×
cover bs=2 / entropic	0.012×	0.0016×	0.00039×	0.25×	0.075×	0.023×
bs=2 below entropic	40/40	40/40	40/40	40/40	40/40	40/40
cover / flow, both bs=2	3.2×	4.2×	5.1×	4.2×	4.0×	3.3×
flow = polymatroid	39	35	30	28	27	30
checked (invalid)	40 (0)	40 (0)	35 (0)	40 (0)	33 (0)	28 (0)
seconds, bs=1/ent/bs=2	0.6/0.5/25	0.8/0.8/57	0.8/2.2/33	0.3/0.3/4.4	0.6/0.8/14	0.9/3.3/22
bs=2, cataloged	0.56	0.007	0.010	—	—	—

exact counts are 10^4–10^9, so this is not a small-count artifact (Table IV). Nor is it a prefix artifact: with each relation replaced by a uniform 20K-row sample of the full file (two fixed seeds) the $|V|$=6 medians are 46/46/49× (cover) and 236/233/244× (polymatroid) for prefix/seed 1/seed 2, all 40 below LpBound in each. Query-level bootstrap 95% intervals (10,000 resamples) keep every median ratio to LpBound in this table and in Table V on one side of 1, except the cover LP on block statistics on AIDS ($[0.42, 1.00]$). None of the 864 bounds with an exact count fails (216 checked queries over both suites × four bounds), nor the 345 three-norm ones (115 Hetionet queries × three). With three norms the entropic baseline is a median 1.9–3.2× looser and the margins correspondingly larger (80/630/2,500×, lower block). Two more rows, with three norms, locate *which* statistic carries the gain. Keeping only the *sizes* of the same pair joins, $|R \bowtie S| = \sum_b \deg_R(b) \deg_S(b)$—the two-relation statistic CorrBound [19] adds to the entropic program, here added exactly (the constraints stay simple, so the flow LP is the polymatroid optimum) though without CorrBound's sketches or multi-relation terms—already beats the entropic bound by a median 15/27/99× (AGM over the same sizes: 7–15×). Under the same program (the flow LP, within 1.73× of the optimum, so these are lower bounds) the interface-conditioned block statistics add a further 19/80/154× (median over all 120: 63×) on 120/120 queries (never below the exact count on the 115 with one).

On Hetionet the price is on the other side of the ledger: bs=2 statistics take 25–57 s at the median (up to 366 s; 33 queries exceeded a 180 s budget, which only flags them: each finished and is kept in every ratio), against under 1 s for bs=1 and 0.5–2.2 s for the entropic LP at these sizes. Those seconds rebuild per query what a system materializes once. The last row measures that: a workload-level catalog keyed by underlying table, orientation and interface holds 676 entries from 91 block builds, costs 527 s in total against the 11,765 s the per-query script spent, and returns the identical bound on 120/120 queries. It saturates by the 68th query, after which a bs=2 estimate costs 7–10 ms, 100–220× below the entropic LP at $|V|$=8–10 and below it on 60/60 of the second half. The 61× collection overhead is paid once per database.

IMDb. The second suite is a different kind of graph (a star schema in which every relation joins movies to people, companies or keywords) and separates the axes only with a program that can use the statistics. With ten norms the inference gap is 5.1–6.8×, and bs=2 statistics are only 1.3/2.9/6.8× tighter than the entropic bound under the cover LP (level with or below that gap, and above the entropic bound on 8 of the 40 queries at $|V|$=6), but 9.7/30/87× under the polymatroid program, on 120/120. On these statistics it beats the cover LP by 7.6–11.6×, against 5.1–6.8× at bs=1: richer statistics reopen the program here too. Under it the bound moves from 23–1,300× the truth to 2.2–15× (101 checked queries, exact counts $3{\cdot}10^5$–$9{\cdot}10^7$ at the median; 19 of the 80 at $|V|$=8–10 timed out; these are the largest, so the ratios to the truth describe the smaller queries). With three norms the entropic

baseline is 3.5–6.7× looser and the cover-LP margin reads 4.0/13/43×. Pair-join sizes barely help (three norms: median 1.00×; they help at all on 24/120, by at most 1.35×): half the pairs share an entity and so join one relation with itself, with size $\sum_x \deg(x)^2$, that relation's ℓ_2^2, which LpBound already has; the movie-sharing pairs are genuine inner products but rarely bind. The interface-conditioned statistics still add 17–171×. (IMDb times were taken under 10 concurrent workers and IMDb has no catalog run; the 61× overhead and the 10 ms cataloged estimate are Hetionet's.) At five times the films (1,000; the 40 queries at $|V|$=6, bounds only, three norms; exploratory) every tightness ratio grows: the inference gap from 3.8 to 4.8×, block statistics from 4.0 to 7.0× under the cover LP and from 17 to 48× under the flow LP, 40/40 below entropic, the program gap from 4.2 to 6.8×. Collecting the statistics grows faster than the data: 4.2× the rows cost 48 s per query at the median (from 4.4 s), up to 25 min and 16 GiB, measured with 10 workers sharing the machine.

G-CARE. As an external check we run every cyclic query of the G-CARE benchmark [15] on its AIDS and YAGO graphs (241 and 366; query sets fixed by G-CARE, size caps below ours), with labelled, directed edges as binary relations filtered by the query's vertex labels, homomorphism counts, and ten norms on both sides (Table V). Our exact counts equal G-CARE's on 241/241 AIDS queries; on YAGO they differ on 49 of the 192 we evaluate, in both directions and under every semantics we tried (directed or not, injective or not; over all 364 cyclic YAGO queries with a count, 80 differ). A second implementation of the same semantics that reads G-CARE's raw files without DuckDB reproduces our count on all 49, and 45 of them lie in G-CARE's Petal families (45 of 100 Petal queries, against 4 of 92 others), which points to those published counts. We therefore check validity against our counts (one YAGO bound, 26, lies below G-CARE's 33 but above our 17). 174 YAGO queries exceed the prototype (a relation over 300K rows or a pair join over 20M) and are reported, not run. Under the polymatroid program, block statistics are 15× (AIDS) and 24× (YAGO) tighter than LpBound and never looser. Under the cover LP the inference gap dominates on AIDS: 59×, as bounded molecular degrees make high norms very informative to the entropic program, against a 1.35× statistics gain (block statistics beat LpBound on only 135/241); on YAGO the two are comparable (6.1× against 8.9×). On both graphs the bs=2 program gap (12×, 2.7×) is below the bs=1 one. G-CARE is thus a regime where the program matters as much as the statistics.

a) Which statistic needs the join: Of the eight interfaces of a block of binary relations $R(a,b), S(b,c)$, seven (five non-trivial) are functions of single-relation degree tables and need only an aggregate join of one relation with the other's degree table, never the pair join: the degree of a is $\sum_{b\in N_R(a)} \deg_S(b)$ and that of b is $\deg_R(b)\deg_S(b)$; only $\{a,c\}$, the common-neighbour count $|N_R(a)\cap N_S(c)|$, needs the join. We compute the seven without it (identical to the materialized statistics on 120/120 Hetionet queries; 0.49 s against 4.6 s per block in our Python prototype). Added first, they close most of the log-gap to the full bound but not the last factor: a median 2.3× (max 18.7×) above the full bs=2 bound on the Hetionet long suite, 1.5–2.9× on the families, 1.9–57× on the sweep. What closes it is one number per block, the ℓ_2 norm of the common-neighbour sequence, $\|A_R^\top A_S\|_F$ for the $b\times a$ and $b\times c$ incidence matrices, whose square counts the 4-cycles a–b–c–b'–a built from two copies of R and of S: a sub-pattern count, where CorrBound's ℓ_1 inner product counts paths a–b–c. Added to bs=1 alone it already reaches a median 1.03× of the full bound (53/120 exact), so the attribution is order-dependent; added to the join-free seven it returns the full bound on 12/12 family–seed pairs, 10/10 sweep points, 85/120 Hetionet and 116/120 IMDb queries (Hetionet median 1.00×, max 1.67×), while its ℓ_∞ does not (1.70×) and its ℓ_1 (the join size, CorrBound's inner product [19]) is already join-free. The one-pass upper bound on that norm from the triangle inequality recovers nothing on Hetionet (2.2×); a 64-probe Hutchinson sketch lands within 0.91–1.78× of the full bound in 0.32 s per block, but is an estimate, not a bound. It does pick which joins to pay for: ranking a $|V|$=6 query's six blocks by how much a sketch of their norm lowers the bound and joining the top three recovers the full $bs\leq 2$ bound on the median Hetionet query (top two on IMDb), against half of that gain for a random three.

A spectral reading. For an even k-cycle of binary relations the count is a trace, $|\mathcal{Q}| = \mathrm{Tr}(A_1\cdots A_k)$ for the incidence matrices A_i ($v_i \times v_{i+1}$), and the common-neighbour ℓ_2 norm is the Frobenius (Schatten-2) norm of the block product $B = A_iA_{i+1}$. Hölder's inequality for Schatten norms gives, over $m = k/2$ consecutive blocks, $|\mathcal{Q}| \leq \prod_j \|B_j\|_{S_m}$, and $\|B\|_{S_m}^m \leq \sigma_{\max}(B)^{m-2}\|B\|_F^2$ with $\sigma_{\max}$ bounded above by a Collatz–Wielandt certificate on $B^\top B$. Two numbers per block, both from the pair join ($\sigma_{\max}$ lies outside the polymatroid catalog), give a bound tighter than the polymatroid program on 96/240 long-suite queries (by a median 1.25× there, though 1.11× looser over all) and never below an exact count; with each block's top 24 singular values (Lanczos; uncertified, yet above every exact count) it is tighter on 220/240 and takes IMDb at $|V|$=10 from 14.9× to 3.2× the truth. The next

TABLE V
G-CARE CYCLIC QUERIES, TEN NORMS ON BOTH SIDES (MEDIANS). YAGO QUERIES WITH A RELATION OVER 300K ROWS (144) OR A PAIR JOIN OVER 20M (30) ARE NOT RUN. EXACT COUNTS BY DUCKDB; G-CARE'S OWN COUNTS AGREE ON 241/241 AIDS AND 143/192 YAGO QUERIES.

	AIDS	YAGO
cyclic queries / evaluated	241 / 241	366 / 192
cover bs=1 / entropic	59×	6.1×
cover bs=2 / entropic	0.74×	0.11×
polymatroid bs=2 / entropic	0.066×	0.041×
cover bs=2 below entropic	135/241	177/192
polymatroid below / equal	241 / 0	188 / 4
entropic / exact	513	2,139
polymatroid bs=2 / exact	66	67
invalid bounds	0	0

step on the statistics axis may thus be spectral order rather than block size. On 22 small random cycles, a search over databases with the same $bs{\leq}2$ statistics puts at most $1.5\times$ of the gap on the program and at least $2.5\times$ on the statistics (cycle-4 medians).

D. Real Data Results

At $bs{=}1$ on real data the LP fires only where cyclic structure meets cross-relation degree asymmetry (Section V-H). The heterogeneous Hetionet cycle-4, four relation types with asymmetric degrees, improves $7.59\times$ (53M$\rightarrow$7.0M, relations capped at 30K edges); the C-D-G and D-G-A Hetionet triangles and the replicated Epinions cycles on 2–5K-edge subsets yield exactly $1.00\times$ (at 20K edges the Epinions cycle-4 improves $2.06\times$, Section V-D). Over six closed metapaths of length 3–5 built from Hetionet's thirteen edge types, LpH-CEG fires *selectively*: it tightens C-G-G ($1.95\times$), C-D-G-G and C-D-A-G-G ($7.59\times$)—exactly those whose hyperpath places the skewed gene–gene relation in a *middle* position—and returns $1.00\times$ on the rest. The effect is *structural*: D-G-G-A holds the same high-skew relation as C-D-G-G yet yields $1.00\times$ because it sits on the closing edge, as Theorem 4 predicts.

Beyond cycles. On a diamond, a 4-clique and a chordal 4-cycle over the same edges LpH-CEG improves 1.9–$4.2\times$ over max-degree (largest on the 4-clique): it fires wherever a skewed relation sits in a middle position, not only on cycles.

A pre-registered suite. A handful of hand-chosen queries cannot support a distributional claim, so we fixed a suite *before* measuring: every type-consistent metagraph cycle of length 3–5 over distinct Hetionet relations (370 queries, admitted by rule), plus seven Epinions shapes. The registry is hash-pinned and the evaluation refuses to run against an edited copy, so the selection could not be changed after measuring. All 377 were evaluated and *none* censored. Over the 370 Hetionet queries the cover/entropic gap has median $1.99\times$, P90 $3.55\times$, maximum $6.29\times$ (215 distinct values), meeting the median $\leq 3\times$, P90 $\leq 8\times$ bar registered in advance. Improvement over max-degree has median $1.47\times$ and reaches $13.1\times$, yet $112/370$ queries (30%) improve by exactly $1.00\times$—the selective behavior above, now over a population rather than a handful. The Epinions half is degenerate and we report it so: one homogeneous relation self-joined makes all seven shapes yield an identical cover/entropic *gap* of $3.28\times$—one distinct observation rather than seven. A second registry extends the rule to lengths 6, 8 and 10, where the 777/4,380/12,960 admissible cycles are too many to solve entropically, so the first 40 per length in SHA-256 order of the canonical relation sequence are admitted (an order blind to data and bounds), giving the 120 Hetionet queries of Table IV. A third registry, fixed after a disclosed pilot, uses IMDb from the Join Order Benchmark [14]: the 200 most-voted feature films (197–37K rows per relation), with one relation for each of nine of the twelve cast roles (costume and production designers and guests omitted when the relations were defined, before any bound was computed), for distributors, for production companies, and for keywords. Each relation joins movies to an entity type of its own, so a person reached as an actor and as a director are different variables: cross-role joins are excluded by design. The schema is then a star—a cycle alternates movies with entities, each entity reached by one relation on both sides, and relations repeat (JOB joins `movie_companies` and `title` twice); of the 364/3,081/25,752 such cycles at length 6/8/10 the first 40 in hash order are admitted. The pilot used the Hetionet rule (one person type, distinct relations) on the first 3,500 feature films of the year 2005; $82/120$ of its joins were empty, so validity could not be tested, and scope and typing were both changed before registering. The pilot favored block statistics more (median cover $bs{=}2$/entropic $9.5{\cdot}10^{-6}$ against 0.077 here); it, and a run at five times the films (Section V-C), are in the artifact.

E. Scalability

On power-law cycles ($|V|{=}100$, $|E|{=}400$, median of 7 runs) the LP adds under 4.7 ms to the max-degree bound (1.17–$1.34\times$ overhead, 8–24 ms in total for $k{=}3$–10) while the improvement grows exponentially with k, reaching $971\times$ at cycle-10.

a) Polynomial inference on block statistics: The Shannon-cone formulation of the entropic bound has $2^{|V|}$ variables and, on the power-law cycles of Figure 2, rises from 81 ms at $k{=}8$ to 25 s at $k{=}12$ and 2,480 s at $k{=}14$ (real Hetionet sequences: 22 s and 27 min), timed end to end with a sparse constraint matrix and HiGHS's interior-point solver. LpBound does not solve that program. Its LPflow [5, Thm. 4.5] follows Im et al. [12]: each *simple* constraint ($|X| \leq 1$) becomes a capacitated edge $X \rightarrow Y$ in a flow network over attribute sets (an ℓ_p constraint adds $\emptyset \rightarrow X$ at capacity δ/p), and the bound is the cheapest capacity assignment that routes one unit of flow to every attribute—a polynomial LP. Our reimplementation returns the entropic bound to 10^{-6} on all 12 instances of Table III and at every $k \leq 14$ here, in 3–26 ms for $k{=}3$–16; the cover LP is 1.25–$1.43\times$ looser than a program that is itself polynomial; both solve in milliseconds.

The question that remains open is inference over *block* statistics, where a $bs{=}2$ block over three attributes contributes constraints conditioned on two of them ($|X|{=}2$)—the non-simple case, which Im et al. show to be as hard as the general polymatroid bound already at $|X| \leq 2$, $|Y| \leq 3$ (their Theorem 1.3(c)). Their §7 relaxation handles such a constraint along a fixed attribute order π by restricting it to the attributes of Y that follow all of X; we carry the ℓ_p term through unchanged. The relaxed constraint is implied by the original (monotonicity), and for ℓ_∞ constraints every order's flow bound is valid by their Theorem 1.5(a); for ℓ_p we have no proof and certify each bound instead. The $2^{|V|}$ polymatroid bound over the *unrelaxed* constraints is valid for every p (each ℓ_p constraint holds on the join's entropy, as in Theorem 1), so a flow bound at or above it is valid. Every real-data flow bound passes (ten norms: $109/109$ short, $240/240$ long-suite, $433/433$ G-CARE and $80/80$ resampled slice queries, equal on 633, to 10^{-6}), as do the synthetic sweep's at $k{=}12$ and 14 (equal; $k{=}16$, where the $2^{|V|}$ program does not finish, is

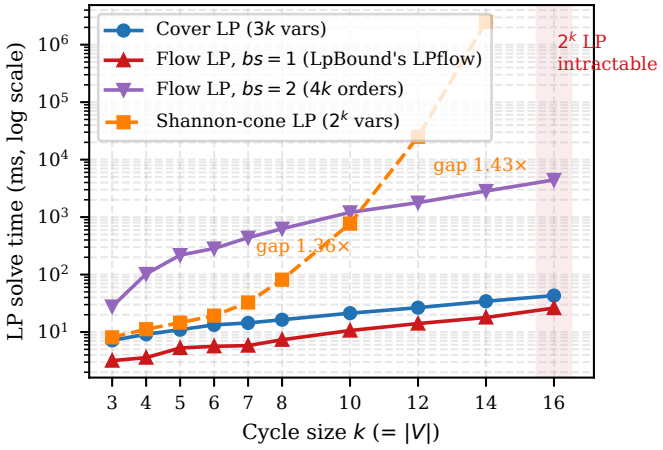


Fig. 2. LP solve time on cycle-k queries (log scale). The Shannon-cone formulation of the bs=1 entropic bound grows super-exponentially (25 s at k=12, 2,480 s at k=14), but LpBound's flow LP reaches the same bound (annotated gap to the cover LP) in 3–26 ms. On bs=2 block statistics the flow LP over $4k$ orders takes 0.03–4.4 s and, with three norms, equals the $2^{|V|}$ polymatroid bound at every even $k \leq 14$.

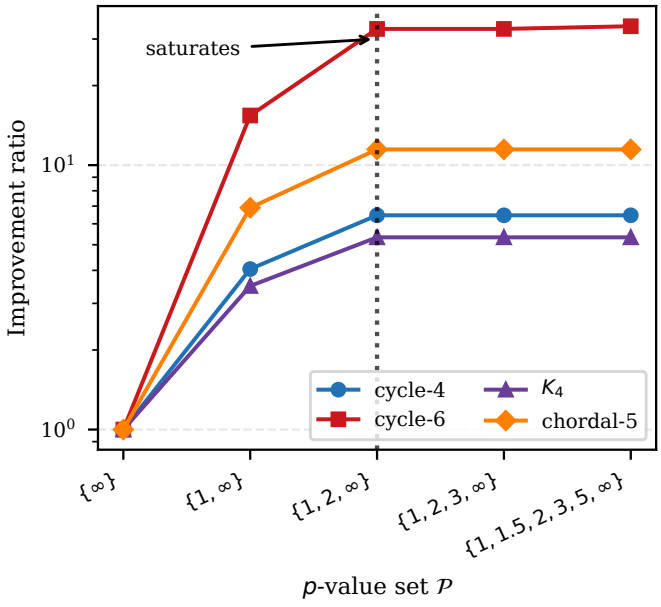


Fig. 3. Sensitivity to the number of p-values across four query families (log scale). Improvement saturates at $\mathcal{P} = \{1, 2, \infty\}$ (dotted line) in every family; additional norms add <2.4%.

uncertified) and 40 random connected shapes ($|V| \leq 5$, all 2,046 order–instance pairs, also never below the exact join). Two facts make the relaxation usable. The $|X|$=2 constraints carry most of the bs=2 gain (dropping them loosens the polymatroid bound 1.5–2.9×), and an order keeps one exactly when both conditioning attributes precede its target—so on a cycle an *alternating* order (one color class, then the other) keeps every constraint centered on the second class; no single order keeps them all, so we solve the flow LP over the $4|V|$ walk and alternating orders and keep the best bound. Over the full bs=2 catalog (every block, every interface) it equals the $2^{|V|}$ polymatroid bound on 12/12 family–seed pairs (three norms; 7/12 with ten) and on every synthetic even cycle up to k=14 (24 min for the $2^{|V|}$ program at k=14) and on the triangle, and loses a constant 1.56–1.59× on cycle-5/7, where one wedge must remain lost; 34/40 of the random shapes are exact (max gap 1.28×). Cost is 4–73 ms per order and 0.03–4.4 s in total at k=3–16 (Figure 2). What it *finds* matters more than what it costs: the cover LP, confined to one hyperpath, leaves most of the block statistics unused, and the flow LP over the same catalog is 1.5–3.8× tighter on the small families (ten norms: 1.6–3.8×) and, with three norms, 2.2× at k=3 growing to 288× at k=16 (uncertified) on the sweep (below the bs=1 optimum by 2–800× where the cover LP manages 1–6×; Figure 1). On real data the $2^{|V|}$ program over the same catalog is cheap enough to solve directly (ten norms: median 0.02/0.18/2.4 s at $|V|$=6/8/10 on Hetionet, at most 63 s at $|V|$=12 on G-CARE), so the ten-norm real-data bounds we report are its optimum. The flow LP equals it on 633 of the 862 reported queries and is within 2.41× on all; it is the route beyond $|V| \approx 12$. On the 120-query Hetionet long suite the median gain of the polymatroid program over the cover LP on the same statistics is 5.0/6.7/11.1× at $|V|$=6/8/10; no bound falls below the exact count.

b) Choice of p-values: On cycle-6, using $\mathcal{P} = \{1, \infty\}$ (2 values) already achieves 17.6× improvement; adding ℓ_2 ($\mathcal{P} = \{1, 2, \infty\}$) reaches 35.7×; further values (ℓ_3, ℓ_5) yield no additional gain (Figure 3). We recommend $\mathcal{P} = \{1, 2, \infty\}$ as the default (cardinality ℓ_1, variance ℓ_2, worst case ℓ_∞); repeating the sweep on four families (cycle-4/6, K_4, chordal-5) confirms this is not cycle-specific—ℓ_2 helps in every family and p beyond $\{1, 2, \infty\}$ adds $< 2.4\%$ to the cover LP. This does not hold for the entropic bound, which gains a median 1.9–6.7× from ten norms on the real suites; comparisons against it therefore use ten norms on both sides (Section V).

F. Bound-Guided Join Ordering

Multi-way ordering. The per-step residuals can order a worst-case-optimal join greedily: GenericJoin iterations drop 2.3× on the Hetionet cycle-4 in our simulator, but over 66 synthetic queries the bound-guided order is 1.3% worse at the median (20 wins, 43 losses; Wilcoxon p=0.003, indicative, as queries share shapes and seeds), so it is a targeted heuristic.

Binary ordering in an engine. Can the cover bound guide *binary* join ordering inside a modern engine, and how does that compare to the engine's own optimizer? We load each query into DuckDB, compute a left-deep order by minimizing the classic C_{out} cost [14] (sum of the bound's estimated intermediate cardinalities), inject it by disabling DuckDB's join-order optimizer, and time real execution. As a reference we also run DuckDB's own optimizer (v1.5.4; its cardinality estimates come from table statistics such as HyperLogLog distinct counts).

Two findings frame the result. First, a cardinality *upper bound* is a conservative cost model: it never underestimates, so ranking plans by it does not beat, and need not even match,

TABLE VI
END-TO-END ROBUSTNESS: SLOWDOWN OVER THE BEST EVALUATED PLAN (DUCKDB, FORCED LEFT-DEEP ORDER VS. THE ENGINE'S OWN OPTIMIZER). LOWER IS BETTER; THE *tail* ROW IS THE WORST CASE OVER ALL QUERIES.

Query	Bound-guided	DuckDB opt.
Cycle-4 (100V/400E)	1.10×	0.60×
Cycle-5 (100V/400E)	1.52×	0.14×
K_4 clique (120V/800E)	1.64×	18.1×
K_4 clique (120V/1500E)	3.32×	21.0×
K_4 clique (120V/2500E)	1.81×	7.2×
K_4 clique (200V/2000E)	1.57×	18.3×
Diamond / chordal-4	2.54×	0.39×
Chordal 5-cycle	2.11×	0.13×
Hetionet △ (real)	3.56×	1.11×
Worst case (tail)	**3.6×**	**21.0×**

an engine optimizer on average; ranking all left-deep orders, the cover LP also correlates less with true intermediate cost than the entropic bound on all four families tried (Kendall τ 0.27–0.87 against 0.33–0.94). Second, and structurally, on a single cycle every proper sub-query is acyclic, where the LP has no cyclic slack (Section III-C), so the two bounds largely induce the same order; the bound changes the ranking only when an *intermediate* sub-query is itself cyclic (cliques, chordal cycles).

What the pessimistic bound *does* buy is robustness against catastrophic plans. Table VI reports, per query, each strategy's slowdown over the best plan we found. DuckDB's optimizer is excellent on average (often faster than any forced left-deep order because it builds bushy plans), but on dense 4-cliques its statistics-based estimates badly underestimate the cyclic intermediate blow-up, and it runs 7–21× slower than optimal across every clique density we tried. Bound-guided ordering, which by construction cannot be lured into such a plan, stays within 3.6× of optimal on *every* query in the suite. The worst-case (tail) slowdown is 3.6× for bound-guided versus 21× for the engine—a 5.9× heavier tail for the engine's optimizer.

a) The tail, over a broader distribution: Of six shapes at two sizes and three seeds, 24 queries survive (cycle-3/4, K_4, diamond; cycle-5 and chordal-5 censored entirely). Medians are close (1.03× vs. 0.93×) while the tails diverge: the bound-guided P95/max are 1.21×/1.31× against DuckDB's 5.82×/8.17× (with n=24, P95 is the second-largest value). Censoring uses the AGM bound, which depends only on relation sizes—censoring by our own bound would drop exactly the queries where it is loose. One caveat: the reference is the best *forced left-deep* order, so bound-guided is $\geq$1 by construction while DuckDB, free to go bushy, scores below 1 on median. This shows a bound-guided choice avoids a plan space's tail, not that it beats the engine.

b) Real data, block statistics: The same protocol on the 40 IMDb cycles at $|V|$=6, with four estimators behind C_{out}, tests the thesis rather than the bs=1 bound. Block statistics move the choice toward the true-cardinality order: the flow LP on bs=2 picks the order exact counts would pick on $24/40$ queries, and on the 33 whose forced orders all finish within 60 s its median slowdown over the best timed order (the strategies' picks plus 16 sampled, of the 96 connected left-deep orders) is 1.19×, against 1.41× for the bs=1 cover LP, 1.23× for LpBound (three norms) and 1.21× for exact counts—differences among the last three within noise. None competes with DuckDB's optimizer here (0.43× median, at most 2.0×): IMDb's cycles do not trigger its tail, and on the other 7 queries almost every forced left-deep order exceeds 60 s while DuckDB finishes, so excluding them favors the forced orders. Each order was timed once while 10 queries ran concurrently, so "within noise" is a judgment, not a test. A better bound improves *which* left-deep plan is chosen, not whether one should be.

LpH-CEG is thus not a replacement for a well-tuned optimizer, but a guide that bounds the worst case on the dense cyclic queries that defeat its estimates (on our synthetic cliques, not on IMDb), consistent with the pessimistic-estimation rationale for worst-case-optimal evaluation [20].

G. JOB, TPC-H, Data Size and Skew (bs=1)

On the JOB-medium subset (CI: 3K, MK: 2K, MC: 1K rows) the two co-actor 4-cycles improve 1.94× (with keywords) and 2.80× (with companies) at bs=1 with weights w=0.5, and a star query by exactly 1.00×; the IMDb suite of Section V-D is the systematic version. On TPC-H (SF=0.01) every query yields 1.00×: dimension keys such as `nation_id` (25 values) are near-uniform, and LpH-CEG needs both *cyclic structure* and *degree skew*. On synthetic cycle-4 (10 seeds) the improvement grows with data size, from 5.9±0.7× at 100 vertices to 33.9±1.5× at 2K (relative std <12%; a single-run sweep reaches 57× at 5K), and peaks at moderate skew: 2.3±0.5× at α=0.5, 16.1±0.8× at α=2.0, 6.6±0.4× at α=3.0. At low skew all ℓ_p-norms nearly coincide, at extreme skew ℓ_∞ is already near-tight, and in between (the range real social and knowledge graphs occupy [21]), ℓ_2 captures the heavy tail.

H. When LpH-CEG Does Not Help

On our workloads LpH-CEG yields no improvement (1.00×) on (1) acyclic star queries (no redundant coverage) and (2) cycles of replicated relations with near-symmetric sampled degrees (the 5K-edge Epinions cycles, Section V-D, and all 8 IMDb co-actor/co-director and replicated movie–actor cycle-4 instances at 5K–1M rows); case (2) is a property of the sample, as 20K edges of the same graph give 2.06× on cycle-4 (Section V-D). What is required is *asymmetric degree structure in cyclic joins*. Synthetic power-law graphs improve more (up to 614× at cycle-8) than real data, where the median over the long suites is 1.8–8.9× (tails 157× on Hetionet, 34× on IMDb): Zipf skew at $\alpha \approx 2$ exceeds real sub-linear degree distributions and one α is shared by all relations.

VI. RELATED WORK

a) Block decomposition methods: CEG's pessimistic variant CEG$_M$ [7], [8] models estimation as a shortest path over single-relation max-degree weights (its optimistic CEG$_O$,

like Graphflow's catalog [22], uses small-join statistics, which we bring into a guaranteed bound); multi-way block extensions [6] generalize this to groups of relations via hyperpaths. Both use only ℓ_∞ statistics. LpH-CEG replaces the scalar weights with LP-optimized ℓ_p-norm products and characterizes *when* the LP can improve: cyclic queries with redundant coverage (Theorem 4).

b) Information-theoretic bounds: The AGM bound [2] uses fractional edge covers with relation cardinalities (ℓ_1-norms). PANDA [3] and MOLP [4] add max-degree constraints (ℓ_∞-norms). LpBound [5] generalizes to all ℓ_p-norms, showing that intermediate norms capture degree skew that max-degree misses. Two results since then bear directly on our reading. Im et al. [12] compute the polymatroid bound for simple degree constraints by a polynomial flow LP and show the non-simple case ($|X|\leq 2$, $|Y|\leq 3$, the shape of a $bs{=}2$ block constraint) as hard as the general polymatroid bound, which is why our flow bound over blocks fixes an attribute order; CorrBound [19] adds inner products of degree vectors across join columns (a two-relation statistic in our terms) to the entropic program and reports 6.7–16× over LpBound on cyclic queries. The first tightens inference at fixed statistics, the second enriches statistics at fixed inference. The ℓ_1-norm of a $bs{=}2$ block sequence *is* CorrBound's inner product (the join size). In a proxy that adds exactly those sizes to the entropic program—optimistic on the pair term (exact sizes, no sketches) and without CorrBound's multi-relation terms, so not comparable to its reported gains—they are worth a median 31× over LpBound on our Hetionet long suite, and the interface-conditioned block statistics a further 63× under the same program. The shared-interface block norms $\sum_b \deg_R(b)^p \deg_S(b)^p$ are ℓ_p-weighted inner products and may fall within CorrBound's vocabulary; the one-side and common-neighbour statistics do not, since they depend on *which* values co-occur, not only how often—and the latter needs the join that CorrBound's sketches avoid (Section V-C). Lin and Wang [23] likewise tighten degree-sequence bounds by counting claw *pairs*, a richer statistic again. SafeBound [24] deploys degree-sequence bounds [25] with predicates on relational workloads, following pessimistic CE [20]; we target graph patterns and a per-step decomposition.

The entropy framework of Abo Khamis et al. [10], under which ℓ_p-norm constraints are information inequalities, is what Theorem 1 applies to block-joined relations.

c) Summaries and learned estimators: COLOR [18] and PathCE [26] build compact summaries; GNCE [17], SPACE [27] and FactorJoin [16] learn estimates. All produce point estimates without a guarantee, for which pessimistic bounds are the safety fallback [6].

d) WCOJ algorithms: Yannakakis+ [28] makes Yannakakis-style evaluation of *acyclic* queries practical in DuckDB and PostgreSQL, the complement of our cyclic setting; CompactLTJ [29] shrinks Leapfrog Triejoin; Capelli et al.'s algorithm [30] is parameterized by prefix upper bounds, which our residuals can supply—though our 66-query study shows they do not beat max-degree ordering in general.

VII. Practical Considerations

a) When to use LpH-CEG: Use it on cyclic queries with $k \geq 4$ and moderately skewed data, with $\mathcal{P} = \{1, 2, \infty\}$; without a cycle, skip the LP. Degree sequences can come from catalog histograms ($\|\mathbf{d}\|_p = (\sum_i c_i d_i^p)^{1/p}$ over buckets (d_i, c_i)) at `ANALYZE` time, on the filtered relation under predicates.

b) Which block size to pay for: At $bs{=}1$, collecting degree sequences on the four cost-study families costs 22–27 ms against ∼1.9 ms to solve; $bs{\geq}2$ multiplies that term by a median 45× there (24× more entries) and 61× (8–558×) on the Hetionet suite, and a cataloged $bs{=}2$ estimate costs 10 ms (Table IV). Use $bs{=}1$ when the bound is re-solved many times per statistics build and $bs{\geq}2$ when the pair joins can be materialized ahead of the workload. Under a selection, block statistics of the unfiltered relations remain valid upper bounds (degrees only shrink) but are looser, and one catalog per predicate is not practical: this, not the program, is the main obstacle to $bs{\geq}2$ under general predicates.

VIII. Conclusion

Replacing a max-degree product with a fractional cover LP is worth a great deal (614× at cycle-8), but on single-relation statistics little is left after it: on a fixed hyperpath the cover LP is optimal over every weighting valid on all databases (Proposition 3), yet sits 1.25–1.43× (synthetic) and a median 2.7–6.8× on Hetionet and IMDb (59× on AIDS) above the entropic optimum, which LpBound's flow LP reaches in polynomial time. With ten norms on both sides, block statistics put the cover LP a median 46–920× below LpBound on registered Hetionet cycles ($|V|{=}6$–10; at $|V|{=}6$ also on random-sample slices), but on IMDb only 1.3–6.8×, level with the inference gap. The polymatroid program over block statistics, valid for every p and solved in seconds here, recovers 9.7–87× there; on Hetionet and IMDb it beats the cover LP by 5.0–11.6× on block statistics against 2.7–6.8× on single-relation ones, so richer statistics reopen the program; the decisive block statistic admits a spectral reading (Section V-C). G-CARE is the counterpoint: 15–24× under the polymatroid program, but under the cover LP the inference gap dominates on AIDS. Our ℓ_p flow LP tracks the polymatroid optimum in polynomial time (equal on 633 of 862 queries), its validity for ℓ_p is certified per query rather than proved, and it is what scales past $|V|{\approx}12$.

Limits. Our evidence is binary graph patterns on Hetionet and IMDb slices (20K rows, 200–1,000 films) and on G-CARE's AIDS and YAGO, where 174 YAGO queries exceed our Python prototype and validity is checked against our own counts, which differ from G-CARE's on 49 of 192. Our LpBound is a reimplementation without its predicate support, and our block statistics are collected without predicates, so STATS-CEB [31] and JOB's own queries are out of scope. Blocks cost a median 61× more to collect, a cost that grows faster than the data and calls for cataloging inside the system. Bound-guided join orders lose to DuckDB on IMDb: this maps where bound headroom lies; it is not an optimizer.

ACKNOWLEDGMENT

Use of AI-generated content. In accordance with the IEEE policy on AI-generated content, we disclose that Claude (Anthropic), a large language model (Claude Opus models, used through the Claude Code tool), was used throughout this work. It drafted and revised text in every section (Sections I–VIII and the abstract); implemented and ran the experiment scripts behind Section V, all tables and figures; wrote the first versions of the proofs of Theorem 1 and Proposition 3 and of the extended version; wrote the automated audit that checks every number in the paper against the source data produced by the released scripts; and produced simulated reviews of drafts. The level of assistance was substantial: the model generated, and did not merely edit, much of the text, code and proofs, under the author's direction. The author directed the work, reviewed the content and takes full responsibility for it.

REFERENCES

[1] M. Abo Khamis, K. Deeds, D. Olteanu, and D. Suciu, "Pessimistic cardinality estimation," *ACM SIGMOD Record*, vol. 53, no. 4, pp. 1–17, 2025.

[2] A. Atserias, M. Grohe, and D. Marx, "Size bounds and query plans for relational joins," in *Proceedings of FOCS*, 2008, pp. 739–748.

[3] M. Abo Khamis, H. Q. Ngo, and D. Suciu, "PANDA: Query evaluation in submodular width," *TheoretiCS*, vol. 4, 2025, preliminary version: "What Do Shannon-type Inequalities, Submodular Width, and Disjunctive Datalog Have to Do with One Another?", PODS 2017.

[4] M. Joglekar and C. Ré, "It's all a matter of degree: Using degree information to optimize multiway joins," in *Proceedings of the International Conference on Database Theory (ICDT)*, 2016, Journal version in Theory Comput. Syst. 62(4), 2018.

[5] H. Zhang, C. Mayer, M. Abo Khamis, D. Olteanu, and D. Suciu, "LpBound: Pessimistic cardinality estimation using ℓ_p-norms of degree sequences," *Proceedings of the ACM on Management of Data*, vol. 3, no. 3, 2025, SIGMOD 2025 Best Paper Award.

[6] S. Lee, "H-CEG: Hypergraph cardinality estimation graphs via multiway block extensions," 2026, under submission.

[7] S. Salihoğlu, J. Chen, Y. Huang, M. Wang, and K. Salem, "Cardinality estimation graphs," *Communications of the ACM*, vol. 69, no. 2, pp. 99–109, 2026, Research Highlight; original: [8].

[8] J. Chen, Y. Huang, M. Wang, S. Salihoglu, and K. Salem, "Accurate summary-based cardinality estimation through the lens of cardinality estimation graphs," *Proc. VLDB Endow.*, vol. 15, no. 8, pp. 1533–1545, 2022.

[9] Q. Huangfu and J. A. J. Hall, "Parallelizing the dual revised simplex method," *Mathematical Programming Computation*, vol. 10, no. 1, pp. 119–142, 2018.

[10] M. Abo Khamis, V. Nakos, D. Olteanu, and D. Suciu, "Join size bounds using ℓ_p-norms on degree sequences," *Proc. ACM Manag. Data*, vol. 2, no. 2, 2024, PODS.

[11] M. Abo Khamis, H. Q. Ngo, and D. Suciu, "Computing join queries with functional dependencies," in *Proceedings of the 35th ACM SIGMOD-SIGACT-SIGAI Symposium on Principles of Database Systems (PODS)*, 2016, pp. 327–342.

[12] S. Im, B. Moseley, H. Q. Ngo, and K. Pruhs, "Efficient algorithms for cardinality estimation and conjunctive query evaluation with simple degree constraints," *Proceedings of the ACM on Management of Data*, vol. 3, no. 2, 2025, PODS 2025; arXiv:2504.02770.

[13] D. S. Himmelstein *et al.*, "Systematic integration of biomedical knowledge prioritizes drugs for repurposing," *eLife*, vol. 6, p. e26726, 2017.

[14] V. Leis, A. Gubichev, A. Mirchev, P. Boncz, A. Kemper, and T. Neumann, "How good are query optimizers, really?" *Proceedings of the VLDB Endowment*, vol. 9, no. 3, pp. 204–215, 2015.

[15] Y. Park, S. Ko, S. S. Bhowmick, K. Kim, K. Hong, and W.-S. Han, "G-CARE: A framework for performance benchmarking of cardinality estimation techniques for subgraph matching," in *Proceedings of the ACM SIGMOD International Conference on Management of Data*, 2020, pp. 1099–1114.

[16] Z. Wu, P. Negi, M. Alizadeh, T. Kraska, and S. Madden, "FactorJoin: A new cardinality estimation framework for join queries," *Proceedings of the ACM on Management of Data*, vol. 1, no. 1, 2023.

[17] T. Schwabe and M. Acosta, "Cardinality estimation over knowledge graphs with embeddings and graph neural networks," *Proc. ACM Manag. Data*, vol. 2, no. 1, 2024, SIGMOD.

[18] K. Deeds, D. Sabale, M. Kayali, and D. Suciu, "COLOR: A framework for applying graph coloring to subgraph cardinality estimation," *Proc. VLDB Endow.*, vol. 18, no. 2, pp. 130–143, 2025.

[19] C. Mayer, H. Zhang, M. Abo Khamis, K. Deeds, D. Olteanu, and D. Suciu, "CorrBound: Cardinality estimation accounting for inter- and intra-relation correlations," *Proc. ACM Manag. Data*, vol. 4, no. 1, 2026, SIGMOD.

[20] W. Cai, M. Balazinska, and D. Suciu, "Pessimistic cardinality estimation: Tighter upper bounds for intermediate join cardinalities," in *Proceedings of the 2019 International Conference on Management of Data (SIGMOD)*, 2019, pp. 18–35.

[21] A.-L. Barabási and R. Albert, "Emergence of scaling in random networks," *Science*, vol. 286, no. 5439, pp. 509–512, 1999.

[22] A. Mhedhbi and S. Salihoglu, "Optimizing subgraph queries by combining binary and worst-case optimal joins," *Proceedings of the VLDB Endowment*, vol. 12, no. 11, pp. 1692–1704, 2019.

[23] Y.-T. Lin and H.-P. Wang, "Ambidextrous degree sequence bounds for pessimistic cardinality estimation," arXiv:2510.04249, 2025.

[24] K. Deeds, D. Suciu, and M. Balazinska, "SafeBound: A practical system for generating cardinality bounds," *Proceedings of the ACM on Management of Data*, vol. 1, no. 1, 2023.

[25] K. Deeds, D. Suciu, M. Balazinska, and W. Cai, "Degree sequence bound for join cardinality estimation," in *26th International Conference on Database Theory (ICDT)*, ser. LIPIcs, vol. 255, 2023, pp. 8:1–8:18.

[26] Z. Wang, Q. Yin, and L. Lai, "Path-centric cardinality estimation for subgraph matching," *Proceedings of the VLDB Endowment*, vol. 18, no. 9, pp. 3063–3076, 2025.

[27] M. Aytimur, T. Chondrogiannis, and M. Grossniklaus, "SPACE: Cardinality estimation for path queries using cardinality-aware sequence-based learning," *Proceedings of the ACM on Management of Data*, vol. 3, no. 3, 2025.

[28] Q. Wang, B. Chen, B. Dai, K. Yi, F. Li, and L. Lin, "Yannakakis+: Practical acyclic query evaluation with theoretical guarantees," *Proceedings of the ACM on Management of Data*, vol. 3, no. 3, 2025.

[29] D. Arroyuelo, F. Campos, A. Gómez-Brandón, G. Linker, G. Navarro, C. Rojas, and D. Vrgoč, "CompactLTJ: Space & time efficient leapfrog triejoin on graph databases," *The VLDB Journal*, vol. 34, no. 6, p. 67, 2025.

[30] F. Capelli, O. Irwin, and S. Salvati, "A simple algorithm for worst case optimal join and sampling," in *28th International Conference on Database Theory (ICDT)*, ser. LIPIcs, vol. 328, 2025, pp. 23:1–23:19.

[31] Y. Han, Z. Wu, P. Wu, R. Zhu, J. Yang, L. W. Tan, K. Zeng, G. Cong, Y. Qin, A. Pfadler, Z. Qian, J. Zhou, J. Li, and B. Cui, "Cardinality estimation in DBMS: A comprehensive benchmark evaluation," *Proceedings of the VLDB Endowment*, vol. 15, no. 4, pp. 752–765, 2021.